

Life history and production of *Agapetus quadratus* (Trichoptera: Glossosomatidae) in a temporary, spring-fed stream

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SUMMARY

1. The life history and trophic basis of production of the caddisfly grazer *Agapetus quadratus* were studied in the torrent Gorg Blau, a spring-fed stream on the island of Majorca that dries annually during summer.
2. Quantitative random samples were taken every 2–3 weeks during an annual surficial flow period, from November 2000 to mid-July 2001. Instars of field-collected larvae were determined by measurements of head width and pronotum length, and the sex of all pupae was determined to study sexual dimorphism and sex ratio.
3. Stage-frequency histograms suggested a trivoltine population, with an average cohort time of 4 months. Larval development was asynchronous, with continuous growth and overlapping generations. Recruitment peaks were identified in mid-November, early March and late June, indicative of winter, spring and summer generations. On average, females were larger than males and the mean sex ratio was 2 : 3 (females : males). Population densities and biomasses derived from the field data were used to calculate production and turnover rate.
4. Annual production of *A. quadratus* in the torrent Gorg Blau ($4.80 \text{ g dry mass m}^{-2} \text{ year}^{-1}$) was the highest ever reported for the genus, being comparable with that estimated for some insects with rapid development and multiple cohorts.
5. Estimates of production of *A. quadratus* were combined with foregut content analysis to estimate the fraction of total production derived from the principal food sources: algae and organic detritus. Algae supported a major proportion of the production of this grazer.
6. The low density of predators characteristic of many temporary streams, and the small amplitudes in discharge and temperature during most of the wet period that characterise the spring habitats might allow high levels of grazer production in this particular Mediterranean stream.

Keywords: caddisfly, grazer, life history, production, temporary stream

Introduction

Ecosystem analyses describing the structure and function of invertebrate communities in streams have frequently used estimates of density and biomass (Vannote *et al.*, 1980; Minshall *et al.*, 1983). However,

because of the central role that benthic invertebrates play in the flow of energy and materials through many benthic freshwater food webs (Wallace & Webster, 1996), measures of secondary production of stream communities are more appropriate to characterise ecosystems in this context (Waters, 1977; Benke, 1984, 1993).

Although the number of studies of secondary production in streams has increased substantially in recent years, few studies have focused on temporary

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ecosystems (see review in Huryn & Wallace, 2000). For instance, Gladden & Smock (1990); Leeper & Taylor (1998), and Whiles, Goldowitz & Charlton (1999) provide some estimates of insect production for intermittent riparian wetlands, and Fisher & Gray (1983) and Jackson & Fisher (1986) others for a lowland Sonoran desert stream. Nevertheless, comparatively little information is available on the ecological characteristics of these systems, and up to date there have been no quantitative studies on life cycles or production of aquatic insects in temporary streams fed by springs.

Estimation of secondary production is accomplished by procedures that depend upon knowledge of various features of the life history of the species involved (Waters, 1979; Benke, 1984), and these may change with environmental conditions. In particular, hydrologic and thermal variability are two of the most important factors influencing life history patterns, especially growth rates and seasonal timing of stream invertebrate life-cycle attributes (Sweeney, 1984; Hauer & Benke, 1987). Upland areas of the island of Majorca are characterised by high predictability and constancy of rain, which assure the replenishment of mountain aquifers that in turn feed the springs that sustain many of the streams of the island. As a consequence, the flow duration of these groundwater, spring-fed streams depends highly on the spring capacity and on the precipitation that fell during the previous years, which directly affects the spring discharge. Although the springs-fed streams draining the island of Majorca generally exhibit low variation in flow and temperature for most of the wet period, the negative hydrologic balance during the warm dry summer that characterises the Mediterranean climate means that, unlike most spring-fed streams elsewhere, these streams are subjected to a summer dry period. Such a range of physical conditions imposed by the annual cyclic events of wetting and drying exert a strong influence on both macroinvertebrate assemblages and ecosystems processes (Dieterich, Anderson & Anderson, 1997; Whiles & Goldowitz, 2001; Álvarez, 2004).

Total density (Delucchi & Peckarsky, 1989; Dieterich & Anderson, 2000) and biomass (Progar & Moldenke, 2002) of aquatic insects in many temporary habitats have been found to be higher than in those with perennial flow. The higher densities and biomass of insects from temporary systems are usually

explained by the capacity of stream fauna living in temporary habitats to avoid desiccation during the dry period (Boulton & Lake, 1992; Lake, 2003) as well as by the low rates of predation that characterise these systems (Progar & Moldenke, 2002). Therefore, in spite of the hydrological disturbance that characterises temporary ecosystems, the secondary production of these systems might also be comparable with that in nearby permanent habitats (e.g. Whiles *et al.*, 1999; Progar & Moldenke, 2002).

Many studies have addressed the importance of food resources to stream consumers by examining their gut contents (e.g. Lamberti & Moore, 1984). Nevertheless, feeding analysis needs to be combined with production estimates in order to understand the ecosystem significance of the study population (Benke & Wallace, 1980; Wallace *et al.*, 1987; Whiles *et al.*, 1999). However, because of practical restrictions, few studies have combined studies of life history, secondary production, and the importance of food resources to the production of single species of stream invertebrates in a way that can be compared among different streams.

The caddisflies in the family Glossosomatidae are frequently dominant components of benthic grazer assemblages inhabiting hard substrata in headwater streams of cool-temperate zones (McAuliffe, 1984; Kohler, 1992; Kuhara, Nakano & Miyasaka, 1999). Larvae of these saddle-case makers graze algae and other materials from the upper surfaces of rocks (Gray & Ward, 1979; Becker, 1990), and constitute a major proportion of total primary consumer production in some streams (e.g. Georgian & Wallace, 1983).

Only one species, *Agapetus quadratus*, is recognised in streams on Majorca. Besides the Balearic Islands, this species is known only from Corsica and, according to Malicky (1992), may be considered as a relict species from former humid times. In Majorca, this species is confined to spring habitats, because the relatively constant conditions of these habitats probably match its physiological requirements (Hynes, 1970). Thus, the existence of this caddisfly species in the spring-fed streams of the island may be an ecological adaptation to the Majorcan climate (Álvarez, 2004).

The first objective of this study was to provide information on the life history of the glossosomatid caddisfly *A. quadratus*, a dominant grazer inhabiting a spring-fed stream on the Mediterranean island of Majorca. A second objective was to assess the

energetic role that this grazer plays in the study temporary stream by estimating secondary production and the trophic basis that determines this production.

Methods

Study site

This study was conducted in the torrent Gorg Blau (GB), a hardwater, spring-fed stream in the Sant Jordi catchment in the mountainous northern area of the island of Majorca (UTM = 499564, 441616). The lower spring reach, which was the study site, is located around 500 m below the springhead and is generally dry during summer. However, the extent of the dry period varies with the rainfall occurring over the year.

A 100-m reach of the stream was sampled immediately before its confluence with the torrent Ternelles. During the study period, torrent Gorg Blau was <1 m wide (mean wetted width over the study period $\pm$ 1SE = 90.9 ± 3.6 cm) and 12 cm deep (10.9 ± 0.6 cm). The dominant habitats were riffles with relatively uniform substrate, mainly of stones and pebbles. Discharge was estimated on each sampling date as the product of cross-sectional area by mean current velocity (Flow Probe, model FP201,

Global Water Instrumentation Inc., Gold River, CA, U.S.A.). Mean flow during the study period was 9.6 L s^{-1} , ranging from 0.4 L s^{-1} in early summer to 31.7 L s^{-1} in winter (Fig. 1b). In the torrent Gorg Blau, distinct increases in the flow levels are infrequent and seasonal, usually occurring in autumn and early spring (Fig. 1b). Water temperature was recorded continuously with a submerged optic Stowaway data logger (Onset Computer, Pocasset, MA, U.S.A.) placed in a riffle area in the middle of the study reach. Although water temperature remained relatively constant during most of the flow period, the study stream showed a larger range of water temperature than most of the spring habitats, reflecting its temporal regime. Mean daily water temperatures remained above 14°C throughout much of the year, but exceeded 20°C at the end of the sampling period (late June and mid-July) (Fig. 1a). Mean water temperature over the study period was 15.5°C , with mean monthly water temperatures ranging from 12.5°C in February to 21.8°C in July.

Sampling procedure

Quantitative samples of *A. quadratus* larvae and pupae were taken approximately every 2–3 weeks during a

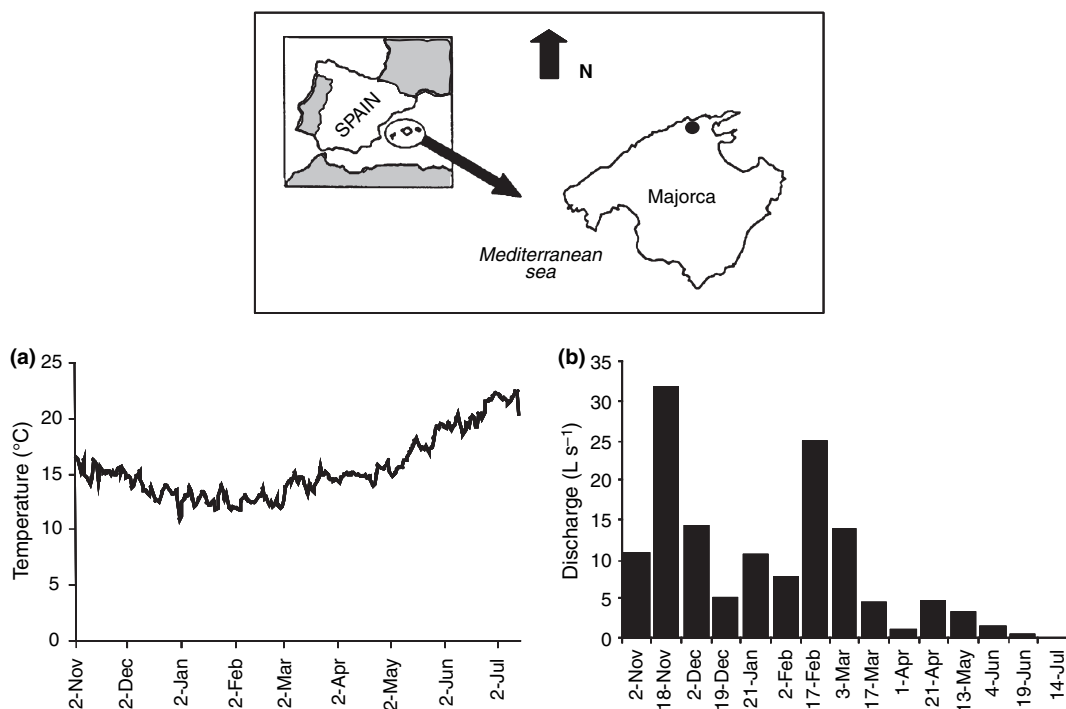


Fig. 1 Location of the study site, mean daily water temperature (a) and discharge (b) on each sampling date at torrent Gorg Blau.

flow period, from 2 November 2000 to 14 July 2001. This period followed a dry summer, and the stream resumed about 1 month before starting this study. A 100-m reach of the torrent was divided into five sections of 20 m and one sample was taken randomly from each section on each occasion. Except for the last sampling date, when the reach was reduced to two isolated, non-flowing pools, samples were always taken from run-riffle habitats of cobble and gravel substrata, which have been shown to be the preferred habitat of Glossosomatidae species (e.g. Wellnitz *et al.*, 2001). A 100- μ m mesh Surber sampler (0.09 m²) sealed around the bottom with sandbags was used to collect the samples. All stones within the sampler frame were scrubbed within the net with a stiff brush to remove attached organisms. Samples were preserved in the field in 10% formaldehyde and their contents were sorted in the laboratory into four size fractions: very coarse (>5 mm), coarse (5–1 mm), fine (1 mm–500 μ m) and very fine (500–100 μ m). *Agapetus quadratus* were picked from the entire coarse fractions (>1 mm) of all samples, but when the volume of sample was too large the fine fractions were subsampled. To do this, samples were evenly distributed on a 500 or 100 μ m sieve (depending of the fraction subsampled) and a fraction of this material (up to one of eight of total) was removed for sorting and counting. *Agapetus quadratus* larvae, prepupae and pupae from each sample were counted and converted to a per square meter basis for each date. Prepupae were defined as larvae found in a resting position in sealed pupal cases, which indicates that feeding had ceased.

Life history parameters

A random subsample of larvae of different sizes (including prepupae) and collected on all dates was used to determine instars by measuring head capsule widths (HW), pronotum lengths (PL) and total body lengths (TL). Body measurements were estimated to the nearest 0.05 mm under an Olympus Bx40 binocular microscope equipped with an ocular micrometer. To determine the number of larval instars, PL was plotted against HW; measures of total length were not used to separate larval instars because there was a high overlap in body length between instars, especially between the latest ones. Measured individuals were dried (1 day at 60°) and weighed to the nearest microgram on a microbalance (Sartorius Micro Scale,

mode M2P, Sartorius AG, Goettingen, Germany). Because of the small size of the first instar, two to four individuals were weighed together, depending on their size. Individual dry mass (M) was plotted against total body length (TL) to calculate a length-mass relationship to predict individual larval biomass. The equation that best described this relationship was: M (μ g) = 7.59 TL (mm)^{2.90} ($r^2 = 0.94$; $n = 1110$; range = 2–12.6 μ g). Before drying, a subsample of the individuals used to obtain this equation was measured to the nearest 0.05 mm with an image analysis system to standardise measurements (manual versus digital) and validate methodology. The microscope was connected to a digital colour camera (Olympus DP10), and the image was read into a PC and processed with image analysis software (Olympus Microimage Software, version 4.0 for Windows, Media Cybernetics, Silver Spring, MD, U.S.A.). Although several body measurements were calculated with the image analysis program (i.e. total length, perimeter and area of the animals) total surface area of the animal was most strongly correlated with the manual measure of total body length (TL) (Pearson's coefficient = 0.96). Therefore, the area of all collected individuals was measured individually by image analysis and converted to body length using the power equation: TL (mm) = 2.27 Area (mm²)^{0.48} ($r^2 = 0.97$) ($n = 180$, range = 0.5–5.0). Then, estimates of TL measured with the image analysis program were used in the dry mass-total length equation to calculate individual mass.

Size frequency distribution of larvae, recruitment (density of first and second instars) and maturation (abundance of prepupal and pupal stages) of the population were calculated for each sampling date in order to estimate the cohort interval.

Exponential curves were used as indices of population dynamics to describe changes in population size through time. The exponential relationship between mean individual mass and time was used to estimate instantaneous growth rate.

To determine the sex ratio at the pupal stage, pupae were sexed by examining the genital primordia. Sexual dimorphism was investigated by drying and weighing a subsample of the pupae collected (170 in total), including approximately 10 of each sex on each date. The length of the larval pronotal sclerite included in the cocoon of each pupa was also measured. The effects of sex on the two ln-transformed variables (mean pupal dry mass and final larval instar pron-

tum length) were determined using one-way ANOVA. Differences in the sex ratio between dates were tested by two-way ANOVA.

Secondary production

Population densities, life history information and biomass estimates derived from the field data were used to calculate secondary production (P) and biomass turnover rate (P/B) of *A. quadratus*. Secondary production was calculated using the size-frequency method (Hynes & Coleman, 1968; Hamilton, 1969; Benke, 1979) and values were corrected for cohort production interval (CPI) (Benke, 1984). The non-cohort method was chosen to calculate production from field data because individuals of all sizes were always present at the study site, making difficult to identify cohorts and to determine patterns of growth accurately. The method calculates a mean size-frequency distribution from samples taken at time intervals and assumes that it represents an average cohort (Benke, 1993). Annual production was calculated for the entire year, using density data from all fitting dates and assuming zero densities for months when the stream was dry. Cohort production was also calculated separately using the size-frequency method for each generation.

Trophic basis of production

Dietary composition of *A. quadratus* was analysed on three occasions: winter (2 February 2001), spring (1 April 2001) and summer (29 June 2001). Foregut contents of late stage larvae were suspended in distilled water, mixed thoroughly and placed onto a microscope slide. Gut contents were divided into two major food categories, algae and amorphous organic detritus, using a microscope (400×) connected to the image analysis system. Inorganic matter was not taken into account, as it is assumed that it is not a component of the animal's diet. The digitised picture was processed with the interactive colour image analysis (Microimage Olympus Software). The proportion of each food category in each gut was calculated using its projected area. The area of five randomly selected microscope fields was estimated from each slide preparation. Average percentages accounted by each food type were calculated from five guts of fourth and fifth instar larvae on each occasion.

Seasonal differences in gut content composition were examined using one-way MANOVA on the percentages of each food item over time. Because the variables were normally distributed (as ascertained by the Kolmogorov–Smirnov test, $P > 0.05$), no transformation was required.

The relative importance of algae and detritus for production of *A. quadratus* was calculated using the trophic basis of production method (Benke & Wallace, 1980, 1997). The amount of *A. quadratus* production attributed to each food type was estimated by multiplying the proportional area occupied by a food type in the foregut by an estimate of the assimilation efficiency for that food category. Assimilation efficiencies of 30% for diatoms and filamentous algae and 10% for amorphous detritus were derived from the literature (e.g. Cummins, 1973; Benke & Wallace, 1980; Slansky & Scriber, 1982). The relative contribution made to production by each food category was then converted to a percentage by dividing it by the sum of relative amounts for the two food types. Net production efficiency (NPE = production/assimilation) was assumed to be 50% (Benke & Wallace, 1980).

All statistical tests were performed with SPSS statistical software (SPSS v.10.1 for Windows, SPSS Institute Inc., Chicago, IL, U.S.A.).

Results

Life history

The relationship between PL and HW indicated that *A. quadratus* has five larval instars (Fig. 2; Table 1).

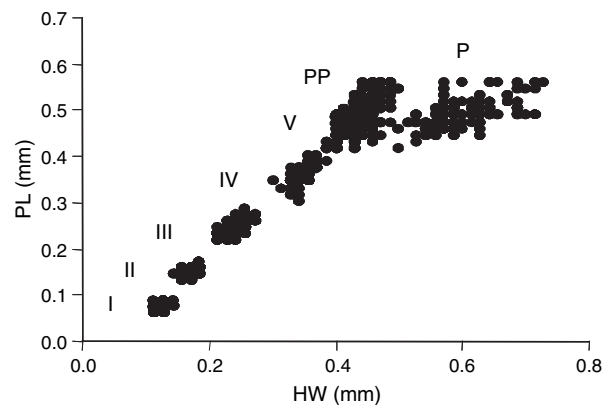


Fig. 2 Relationship between pronotum length (mm) and head width (mm) of *A. quadratus* Larvae (instars I–V), prepupae (PP) and pupae (P).

Table 1 Dry weights and pronotum lengths of larval instars I–V, prepupae and male and female pupae of *A. quadratus*. *n* = the number of individuals weighed and measured.

Stage	<i>n</i>	Dry mass (µg)		Pronotum length (mm)	
		Mean	Range	Mean	Range
I	1252	3.0	1.3–4.7	0.07	0.01–0.12
II	1491	8.8	4.7–14.4	0.17	0.12–0.22
III	2040	27.7	14.5–44.3	0.27	0.22–0.32
IV	3284	82.1	44.4–128.9	0.37	0.32–0.41
V	3071	229.2	128.9–887.4	0.46	0.42–0.59
Prepupae	553	293.3	90.8–677.4	0.48	0.37–0.56
P-female	132	397.9	129.0–1217.0	0.54	0.49–0.73
P-male	140	225.6	89.0–542.0	0.49	0.48–0.71

Life stage frequency histograms showed that in the torrent Gorg Blau *A. quadratus* had a trivoltine life history, with an average cohort time of 4 months (Fig. 3). Nevertheless, hatching and pupation continued at a low rate throughout the year, indicating a degree of asynchronous larval development (Fig. 3). Few prepupae were recorded on all dates, thus this stage is of short duration, whereas pupae were always present at relatively higher densities, reflecting a prolonged pupation period (Fig. 3). Peaks in prepupal and pupal numbers were apparent in mid-November, early March and early June. These maturation peaks were followed by three recruitment peaks in early December, late March and late June. At the times of these recruitment peaks, first and second larval instars

accounted on average for 67.2% of all larvae collected (ranging from 48.6% in spring to 82.3% in winter; Fig. 3).

Population growth of *A. quadratus* (increase in larval density) was significantly explained by an exponential curve, and indicated an instantaneous rate of natural population increase (*r*) of 0.022 day⁻¹ (*r*² = 0.64, *P* < 0.01). Fluctuations in population densities corresponded with periods of larva recruitment and the three decreases were synchronised with the maturation periods (Fig. 4a). Mean density for each generation (mean population size ± 1 SE) was 2614 ± 507 individuals m⁻² for the winter generation, 7280 ± 2467 individuals m⁻² for the spring generation and 13915 ± 7934 individuals m⁻² for the summer generation.

Changes in mean individual mass (*M*) over time clearly identified three periods of constant exponential growth: the first from 2 December to 3 March (winter generation), the second from 17 March to 4 June (spring generation) and the third from 29 June to 18 November (summer generation) (Fig. 4b). Within these periods, mean daily instantaneous growth rate was estimated using an exponential relationship between the log-transformed values of the mean individual dry mass for each date over time (in days) (Table 2). Daily instantaneous growth rates for each generation (*G* = % dry mass day⁻¹) were calculated as:

$$G = 100 \times g; \quad g = \frac{(\ln M_m - \ln M_r)}{t}$$

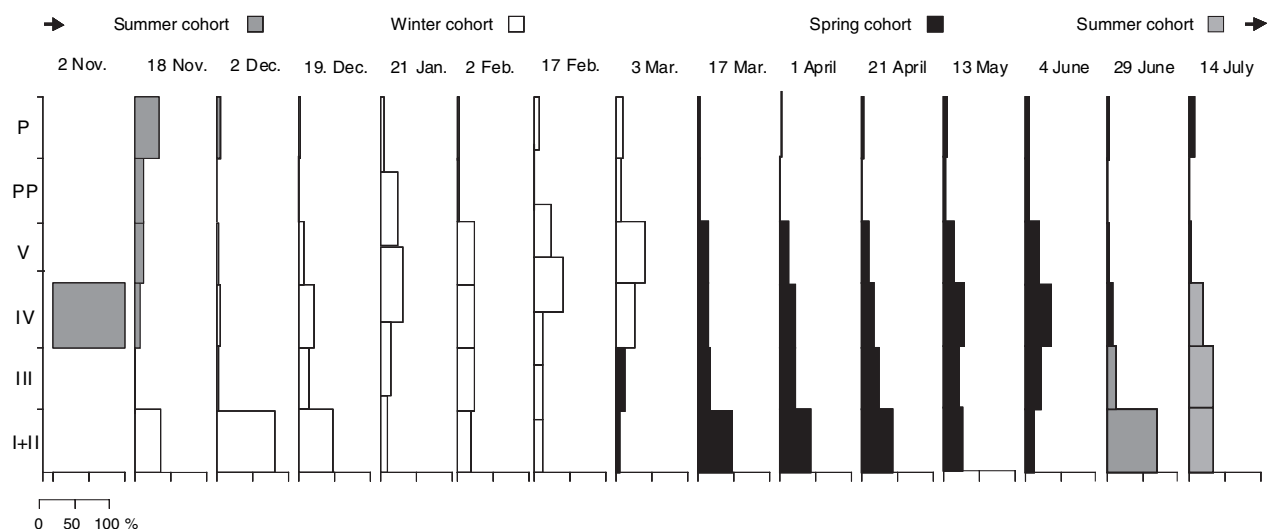


Fig. 3 Relative abundance (%) of larval instars I–V, prepupae (PP) and pupae (P) of *A. quadratus* on each sampling date. The sum of all bars on a sampling date is 100%.

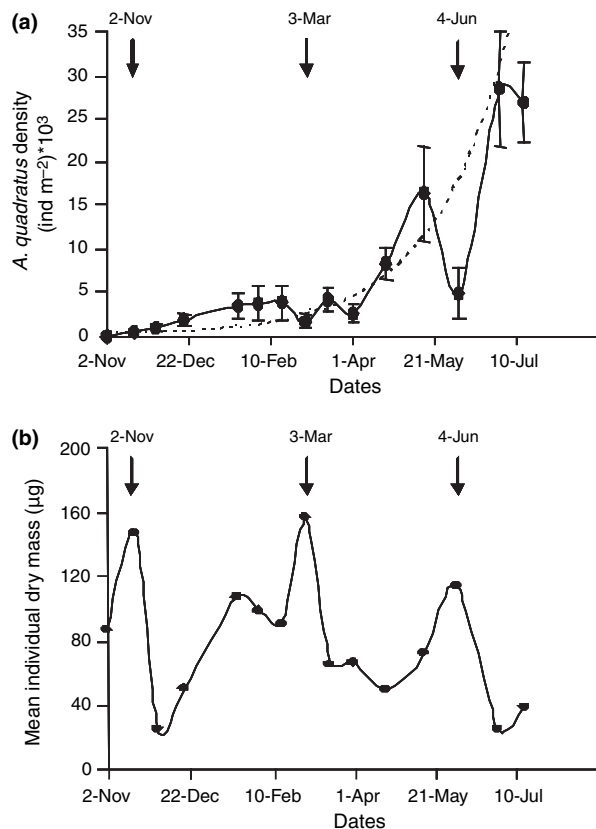


Fig. 4 Mean density (larvae and pupae) ($\pm$ SE) (a) and mean individual mass (b) of *A. quadratus* over time. Arrows indicate the beginning of each generation. (a) shows the exponential curve adjusted to the densities estimated on each day.

where M_m , individual dry mass at the maturation peak (158 μ g on 3 March, 116 μ g on 4 June and 148 μ g on 18 November) (Fig. 4b), M_r = individual dry mass at the recruitment peak (25 μ g on 17 March, 65 μ g on 29 June and 25 μ g on 2 December) and t = duration of each generation (Table 2). Larvae grew faster in winter (1.74% dry mass day⁻¹) than in spring (0.62% dry mass day⁻¹). The third generation hatched at the end of June and grew slowly (1.06% dry mass days⁻¹) until emerging in mid November (Fig. 4b, Table 2).

Table 2 Mean instantaneous growth (g, % dry mass day⁻¹) of larvae and prepupae of *A. quadratus* during three periods (cohorts); n = number of samples (dates) used for the estimation. See text for the mixture of summer 2000 and summer 2001 data.

Cohort	Period	t (days)	g (% dry mass)	r^2	P -value	n
Winter	2 December to 3 March	105	1.74	0.86	<0.05	6
Spring	17 March to 4 June	93	0.62	0.45	>0.05	5
Summer	29 June to 18 November	167	1.06	0.95	<0.05	4

One-way ANOVA indicated that mean dry mass of female pupae was significantly higher than males ($F_{1,229} = 104.83$, $P < 0.001$) (Table 1). Similarly, the 1-way ANOVA on the PL taken from pupal cocoons indicated that sexual dimorphism was apparent in the final larval instar or prepupae (one-way ANOVA $F_{1,147} = 90.31$, $P < 0.001$) (Table 1), as frequently occurs in caddisflies. Moreover, mean dry mass of both sexes gradually decreased during the year, with the highest dry masses found at the beginning of the sampling period and the lowest at the end.

Mean percentages of male and female pupae over the studied period were significantly different (two-way ANOVA $F_{1,112} = 18.84$, $P < 0.001$). The mean percentage of females was 37.1%, and ranged from 9.1 to 59.3% in different months. For males it varied monthly from 40.7 to 90.9%, with a mean percentage of 62.9% over the studied period. The overall sex ratio of pupae was therefore 2 : 3 in favour of males. Although fluctuations in sex ratio occurred throughout the study period, significantly more male than female pupae were recorded (two-way ANOVA sex $\times$ date $F_{4,12} = 3.79$, $P = 0.027$).

Production

Annual production of *A. quadratus* was estimated to be 4.80 g dry mass m⁻² year⁻¹, which corresponds to an annual P/B ratio of 11.0 and a cohort P/B of 3.6 (Table 3). Cohort production of *A. quadratus*, estimated from the cohorts identified in Table 2, was 1.28 g m⁻² for the winter generation (P/B ratio = 3.1), 2.57 g m⁻² for the spring generation (P/B ratio = 3.6), and 1.0 g m⁻² for the summer generation (P/B ratio = 4.6).

The two major food types consumed by *A. quadratus* were algae (green algae and diatoms) [mean 70.3 $\pm$ 2.5% (SE)] and amorphous detritus (29.7 $\pm$ 2.5%; Table 4). No differences in the proportion of each food category ingested were found over time (one-way MANOVA $F_{4,156} = 2.77$, $P > 0.05$). Algae

Table 3 Annual biomass and production of *A. quadratus* (all cohorts combined) using the size-frequency method. All units are in mg dry mass.

Size class	Average cohort N	N (no. m ⁻²)	ΔN (no. m ⁻²)	Individual mass (W; mg)	Mass at loss (mg) [$W_m(W_1+W_2)/2$]	Biomass (mg m ⁻²)	$W_m\Delta N$ (g m ⁻² year ⁻¹)	$W_m\Delta N \times N_s(5)$ (uncorrected P) (g m ⁻² year ⁻¹)	Corrected P ($\times 3$ cohorts) (g m ⁻² year ⁻¹)
I	626.8	1503.2	-591.3	0.003	0.006	4.27	-0.004	-0.018	(0.055)
II	873.4	2094.5	569.9	0.009	0.019	19.49	0.011	0.053	0.162
III	635.8	1524.6	140.9	0.028	0.056	42.64	0.008	0.040	0.121
IV	577.0	1383.7	283.4	0.085	0.157	116.96	0.045	0.223	0.677
V	458.8	1100.3	1100.3	0.230	0.230	252.77	0.253	1.264	3.844
Total		7606.3				436.1		1.56	4.80

Annual P/B [corrected P/B (g)] = 11.0.

Cohort P/B [uncorrected P/B (g)] = 3.6.

accounted for a large fraction (87.5%) of the production of *A. quadratus* (Table 4). Total annual production of the *A. quadratus* population was estimated to require the consumption of 28.0 g m⁻² of algae and 11.9 g m⁻² of amorphous detritus (Table 4). The total amount of food consumed by the population of *A. quadratus* would therefore be 39.9 g m⁻² with about 30.3 g m⁻² egested as faecal material (Table 4).

Discussion

Life history

Size frequency distribution of *A. quadratus* indicated the presence of five larval instars, as occurs in most Trichoptera, including glossosomatids (e.g. Anderson & Bourne, 1974; Georgian & Wallace, 1983). In this study *A. quadratus* exhibited a trivoltine life history, a situation that has not been reported previously for species of *Agapetus* (see Table 5). However, Jin (1994) found that *Glossosoma nigror* had three generations per year in two Alabama streams, and attributed this voltinism to the effect of high water temperature, which averaged around 15 °C in both streams. In the Majorcan spring-fed stream, temperatures were relatively high throughout the year, being over 10–15 °C even on the coldest winter days. In contrast with the synchrony showed by *Agapetus* in most streams, an asynchronous development was found for *A. quadratus* in this study. Iversen (1976) reported a similar asynchronous life history for *A. fuscipes* in a spring in Denmark. In fact, several authors have suggested that the asynchrony in the life cycles of some aquatic insects in spring habitats should be attributed to the constant temperature regime, and therefore to the lack of temperature cues, that

characterised these systems (Dobrin & Giberson, 2003).

Growth. In mid-July, the stream reach was reduced to two isolated pools and grazers were highly concentrated. Population size subsequently decreased on 14 July, before the water stopped flowing. Increasingly harsh environmental conditions (low water level, high water temperature and low oxygen concentration) and limited resources for growth and reproduction (space or food availability) are likely contributing factors.

The low density of larvae at the beginning of the sampling period (2 November) suggests that females of *A. quadratus* that survived through summer, may lay the eggs in the more permanent upper reaches of the stream. These areas, closer to the spring source, are most likely to retain surface water for egg survivorship or larval development, and provide a reservoir of individuals that are able to recolonise the lower parts of the stream once it starts flowing. This pattern of migration between temporary and permanent areas is similar to that observed for other caddisfly populations that inhabit headwater springs (Meyer & Meyer, 2000) and temporary wetlands (Wissinger, Brown & Jannot, 2003).

The mean daily growth rate of *A. quadratus* during the winter generation (1.74% dry mass day⁻¹) fell within the range of values reported by Marchant & Hehir (1999) for *A. monticulus* (1.8–1.9% dry mass day⁻¹) in Little River, Australia, but is lower than that reported for a rapidly growing North American species of *Agapetus*, in which growth rates reached 5.9% dry mass day⁻¹ (Georgian & Wallace, 1983). Although the pattern of individual growth was

Table 4 Diet analysis, fraction of production attributed to each food source. The amount of food consumed and the amount of faeces produced by *A. quadratus*.

Food type	Food type in foregut (%)	Assimilation efficiency (AE)*	Net production efficiency (NPE)*	Gross production efficiency (AE × NPE)*	Amount of food attributed to production (%)	Production attributed to each food type (%)	Production from each food type (g m ⁻² year ⁻¹)	Food consumed (FC) (g m ⁻² year ⁻¹)	Faeces produced (g m ⁻² year ⁻¹)
Algae	70	0.3	0.5	0.15	10.5	87.7	4.21	28.05	19.63
Detritus	30	0.1	0.5	0.05	1.5	12.3	0.59	11.85	10.67
Total				0.2	12.0		4.8	39.9	30.3

*Values derived from literature; see text.

Annual production estimated for *A. quadratus* (g m⁻² year⁻¹) = 4.80; see Table 3.

broadly similar throughout the year, it was highest for the winter generation and lowest for the spring generation. In fact, besides having the lowest daily instantaneous growth rate, larvae from the spring generation also exhibited the smallest individual maximum size attained. Growth rates of benthic invertebrates are usually strongly related to water temperature (Benke, 1993; Marchant & Hahir, 1999), thus, the faster population growth of the spring generation (i.e. higher number of individuals per day) together with the increased metabolism imposed by the high temperatures, may have limited individual growth rates, and therefore may explain the observed smaller size of individuals from this generation (Vannote & Sweeney, 1980).

The presence of late instar larvae in November indicates either that during summer the torrent Gorg Blau dried out for only a short period, or that *A. quadratus* may be adapted to withstand a longer dry period. In fact, a common characteristic of some insects inhabiting temporary habitats is rapid larval development; a strategy that allows them to complete their development before dry periods occur (Wissinger, Brown & Jannot, 2003), or diapausing eggs that allow insects to avoid temperature extremes and dry conditions (Williams, 1987; Delucchi & Peckarsky, 1989). However, the growth rate of *A. quadratus* observed in Gorg Blau was similar to that of other *Agapetus* in temperate streams. In addition, no summer diapausing eggs have been reported for this genus. In contrast, *A. bifidus* over-wintered in the egg stage in a warm Oregon brook (Anderson & Bourne, 1974), and larval diapause was reported for *A. yasensis*, which over-wintered as third and fourth instars in a cold stream in Japan (Sameshima & Hiroaki, 1994) (Table 5).

Sexual dimorphism and sex ratio. A distinct sexual dimorphism was apparent in the pupae of this species, as observed for other *Agapetus* (Becker, 1987). As most caddisflies do not feed as adults, female size is a good indicator of fecundity (e.g. Sweeney & Vannote, 1981). Thus in those cases in which females are larger than males, eggs may account for much of the weight difference (Butler, 1984). However, as indicated by the difference in size observed in the pronotums taken from pupal cocoons, size dimorphism of *A. quadratus* was already apparent in final instar larvae, as showed for other insects (e.g.

Table 5 Review of previous studies on the life cycle and production of *Agapetus* spp. Water temperature amplitudes are daily mean values.

Reference	Species	Larval instars	Voltinism (CPI)	Annual production (g m ² year ⁻¹)	Habitat/location
Anderson & Bourne (1974)	<i>Agapetus bifidus</i>	5	Univoltine (5 months)*	n/a	Temp. Stream/W USA
Anderson (1973)	<i>Agapetus fuscipes</i>	7	Univoltine	n/a	Temp. Stream/W USA
Castro (1975)	<i>Agapetus fuscipes</i>	8	Univoltine	0.62	Spring/Denmark
Iversen (1976, 1988)	<i>Agapetus fuscipes</i>	8	Univoltine (1 year)	0.08	Spring/Denmark
Recaséns & Murillo (1986)	<i>Agapetus fuscipes</i>	7	Bivoltine	n/a	Stream/NE Spain
Bowles & Allen (1992)	<i>Agapetus illini</i>	5	Univoltine	n/a	Stream/SW USA
Gawne (1993)	<i>Agapetus kimminsi</i>	n/a	Univoltine	n/a	Stream/SE Australia
Dean & Cartwright (1987)	<i>Agapetus monticolus</i>	n/a	Univoltine	n/a	Stream/SE Australia
Marchant & Hehir (1999)	<i>Agapetus monticolus</i>	5	Univoltine (10 months)	0.04 [†]	Stream/SE Australia
Neves (1979)	<i>Agapetus pinatus</i>	n/a	n/a	0.002	Stream/NE USA
Marchant & Hehir (1999)	<i>Agapetus pontona</i>	5	Univoltine (5–6 months)	0.10 [‡]	Stream/SE Australia
Álvarez & Pardo (present study)	<i>Agapetus quadratus</i>	5	Trivoltine (4 months)	4.91	Tem. Stream/E Spain
Sameshima & Hiroaki (1994)	<i>Agapetus yasensis</i>	n/a	Univoltine (1 y)*	n/a	Stream/Japan
Georgian & Wallace (1983)	<i>Agapetus</i> sp.	5	Univoltin (44 d)	0.02	Stream/E USA

*Winter diapause.

[†]Estimated for 3 months.[‡]Estimated for 5–6 months.

n/a, not available.

Stenoperla prasina, Plecoptera, Winterbourn, 1974; *Chironomus tardus*, Diptera, Butler, 1982; and *Zelandopsycha ingens* or *Odontocerum albicorne*, Trichoptera, Winterbourn & Davis, 1976; Elliott, 1982).

A deviation in the mean sex ratio of pupae suggests that more males than females pupated and emerged during the year. A similar preponderance of males was also reported for *O. albicorne* in a Lake District stream (Elliott, 1982), whereas Marchant & Hehir (1999) reported a 1 : 1 sex ratio for pupae of two species of *Agapetus* in a south-eastern Australian stream. In fact, this 1 : 1 sex ratio is the most common ratio reported for caddisflies (e.g. Wiggins, 1996). Counting pupae ensured that the sex ratios were not biased by adult behaviour which, in some cases, can result in unrepresentative collections of the sexes (Butler, 1984). Some studies indicate that deviations from a 1 : 1 adult sex ratio may be a consequence of differences in larval mortality of males and females prior to emergence (e.g. LeSage & Harrison, 1980) rather than a genetic sex-determining mechanism. For example, Elliott (1982) suggested that the larger size of caddisflies females at emergence may require higher energy reserves and hence higher level of foraging activity, which may increase their vulnerability to predation prior to pupation. In fact, as with most Trichoptera, up to 80% of the total larval mass (at the prepupal stage) was gained in the final larval stage,

during which significant lipid reserves are accumulated prior to pupation (Becker, 1987, 1990).

Production

Estimates of secondary production of *A. quadratus* were at least an order of magnitude higher than values published for other *Agapetus* species (Table 5). Thus, the value of 4.80 g dry mass m⁻² estimated for *A. quadratus* in the torrent Gorg Blau contrasts with estimates between 0.002 g m⁻² for *A. pinatus* in streams of Massachusetts (Neves, 1979) and 0.62 g dry mass m⁻² for *A. fuscipes* in Germany (Castro, 1975). Marchant & Hehir (1999), estimated that annual production of the univoltine *A. pontona* ranged from 0.03 to 0.22 g m⁻² in a south-eastern Australian stream, and Georgian & Wallace (1983), recorded a value of 0.02 g m⁻² for an unidentified species of *Agapetus* in a southern Appalachian stream (Table 5).

Agapetus quadratus exhibited a continuous exponential population development, which might reflect the favourable conditions of the habitat: a predator-free environment where food supply is not limiting (Álvarez, 2004). Another factor that may promote the high production of *A. quadratus* in the island is the absence of other competing grazers (Progar & Moldenke, 2002; Álvarez, 2004) that make it more likely for this caddisfly species to be highly produc-

tive in the studied stream. In fact, in many streams in which *Agapetus* was poorly represented there were several other grazers (e.g. Georgian & Wallace, 1983). Similarly, Whiles *et al.* (1999) attributed the high values of production obtained for a small shredder limnephilid in an intermittent wetland, to the high water temperatures, the high abundance of food and the low density of other competitors that characterised the studied temporary systems.

Values of cohort P/B obtained in this study (3.6) fall at the lower end of the range of variation for temperate stream invertebrates (usually 2 to 8) (Waters, 1977; Benke, 1984). However, the annual P/B estimated (annual P/B = 11.0) was particularly high compared with the P/B ratios observed in other univoltine *Agapetus* species (e.g. Georgian & Wallace, 1983), which may derive from the trivoltine life history that characterise this species. In fact, the high annual production reached for *A. quadratus* might mainly come from the high rates of annual P/B, rather than from the high values of biomass accrual or rapid growth alone, as occurs in other insects (e.g. Iversen, 1988). Small size at maturity is characteristic of many insects with short generation times and multiple cohorts (Fisher & Gray, 1983; Benke, 1984; Jackson & Fisher, 1986) and may be an adaptation to the high temperatures (Vannote & Sweeney, 1980) and to the disturbance derived from the periodic drying and wetting that characterises temporary streams (Gray, 1981).

Trophic basis of production

Previous studies on feeding preferences of grazing glossosomatids have also shown that epilithon (algae and detritus) is the major component of their diet (e.g. Anderson & Bourne, 1974; Becker, 1990), although the proportion of each food category can be highly variable. For example, Cummins (1973) calculated algal proportions of 26 and 98% in the guts of larvae of *Glossosoma nigrum* in a forested and in an open-canopy stream, respectively. However, algae accounted only 15–25% of the gut contents of *Agapetus* spp. In upland streams (Becker, 1990; Gawne, 1993) while organic detritus made up the remainder of gut contents. Such low values did not occur in the present study.

Many benthic macroinvertebrates with short generation times, high reproductive rates and long

periods of recruitment, have a strong probability that their production will track their food supply (Wallace *et al.*, 1999; Melody & Richardson, 2004). Algae accounted for 87.5% of *A. quadratus* annual production, which compares well with the 75% of glossosomatid production attributed to the consumption of diatoms in a forested headwater stream (Mayer & Likens, 1987). Assuming that the literature-based assimilation efficiencies used in the analysis were accurate, the population of *A. quadratus* would require an annual consumption of approximately 30 g m⁻² of algae and approximately 12 g m⁻² of amorphous detritus to support its production. In the torrent Gorg Blau, mean values of epilithic organic matter measured for the same period as in the present study were around 8 g dry mass m⁻² (Álvarez, 2004), which is approximately 20% of the consumption requirement of *A. quadratus*. However, in an adjacent stream where this species was rare (T5) mean values of epilithon for the studied period were 37 g dry mass m⁻² (Álvarez, 2004), which agrees with the consumption requirement of *A. quadratus*. Therefore, large grazer standing crops in the torrent Gorg Blau may be supported by rapid algal turnover rate, which may be easily accomplished by benthic algae in this shallow stream.

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